

Architectural Particular Specification

WATERPROOF CONCRETE

SECTION 2.2

Rev	Date	Description
0	14 APR 2022	First Issue

Main Contract for
Proposed Development at 2-16 Lam Tin Street, Kwai Chung, New Territories
Lot No. 302 in D.D. 444
Andrew Lee King Fun & Associates Architects Limited

SECTION 2.1

WATERPROOF CONCRETE

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The scope of work shall consist of design, selection, supply, laying, installing, assembly, protection and testing of waterproofing concrete and waterstop for non-moving construction joint for all areas as specified in the contract specifications and drawings, which including but not limited to:
1. Waterproof concrete for basement concrete screen wall, lift shaft, water tank etc. and in all other areas so designated on the Drawings, etc.
 2. The details of waterproof concrete and waterstop shall be read in conjunction with architectural particular specification Section 6.1 and relevant structural engineer's specification.
 3. Consulting Engineers' Contract Documents.

1.02 REFERENCES

- A. British Standards (BS):
1. BS 3797:1990 - Specification for Light Weight Aggregates for Masonry Units and Structural Concrete.
 2. BS4449: 1988 - Specification for Carbon Steel Bars for the Reinforcement of Concrete.
 3. BS6319: All applicable parts : Test of Resin and Polymer/Cement Compositions for Use in Construction (date varies).
 4. BS 8000: Part 9: 1989 - Code of Practice for Cement/Sand Floor Screeds and Concrete Floor Toppings
 5. CP 204 : Part 2 : 1970 - In-situ Floor Finishes
- B. American Concrete Institute (ACI) :
1. ACI 301: Specifications for Structural Concrete for Buildings.
 2. ACI 305R: Hot Weather Concreting.
 3. ACI 306R: Cold Weather Concreting.
 4. ACI 308: Standard Practice for Curing Concrete.
 5. ACI 315 : Details and Detailing of Concrete Reinforcement.
- C. American Society for Testing and Materials (ASTM):
1. ASTM A185 : Welded Steel Wired Fabric for Concrete Reinforcement.
 2. ASTM A615: Deformed and Plain Billet - Steel Bars for Concrete Reinforcement.
 3. ASTM C31 - 91: Making and Curing Concrete Test Specimens in the Field.
 4. ASTM C33 - 93: Concrete Aggregates.
 5. ASTM 39 - 93a: Compressive Strength of Cylindrical Concrete Specimens.
 6. ASTM C94 - 94: Specification for Ready-Mixed Concrete Specimens.
 7. ASTM C143 - 90a: Test Method for Slump of Portland Cement Concrete.

8. ASTM C150 - 94: Specification for Portland Cement.
9. ASTM C172 - 90: Method of Sampling Freshly Mixed Concrete.
10. ASTM C881 - Epoxy Resin Base Bonding Systems for Concrete.

1.03 QUALITY ASSURANCE

(NOUT USED)

1.04 PERFORMANCE REQUIREMENTS

Corrosion-proof and waterproof concrete for superstructure shall comply with the conditions and requirements as set out in this section.

- A. A kicker from the final finish level for the waterproof concrete slab/pile cap are to cast monolithically for the future construction of R.C. screen wall to achieve an integral waterproofed tanking for the whole basement structure. The kicker is under foundation contract as indicated in architectural drawings. Both waterproof concrete and basement tanking systems carried out by Foundation Contractor and the Main Contractor must be compatible with each other.
- B. All flat roofs and parapet wall are to be cast with crystalline type waterproof concrete and a buffer zone either an extension up or down, or a slab beyond for 300mm shall be form monolithically at the edges of the slab or parapet wall.
- C. The Contractor shall note that there may be parts of shoring and struts erected during any basement or other excavation that may be embedded in basement floors or walls permanently. The Contractor shall provide any additional attachment or projection to the shoring to be embedded to ensure the water-tightness at these joints shall not be reduced.
- D. The Main Contractor shall be responsible for carrying out the construction of the specify areas, and other designated areas to minimize the adverse effect of shrinkage to ensure watertight concrete work. Waterproof concrete for the Basement Walls shall not be cast directly against any piles with sectional and restraint change e.g. sheet piles. Where sheet piles are present, the Main Contractor shall erect formwork across the sheet piles, so as to provide a flat back for the Basement Walls, prior to the casting of waterproof concrete. Construction sequence shall be agreed by the Supplier of liquid applied crystalline plasticizer and submitted to the Architect/Engineer.
- E. Treatment of tie-bolts, shoring strut and other embedment in corrosion-proof and waterproof Concrete, and Remedial Work for honeycomb and crack in corrosion-proof and waterproof Concrete, will be applied by the liquid applied crystalline plasticizer waterproof concrete system supplier. Tenderers will be reminded to include their normal contingency for defects repairs in their tenders. It shall be expressly agreed that proper compaction and curing of concrete is the responsibility of the concreter, and costs for repair of honeycombs and cracks are chargeable to the Main Contractor.
- F. All waterproof concrete shall be mixed with liquid applied "SCHOMBURG Betocrete CL-170-P" (Formerly known as "SCHOMBURG Betocrete C17(BV)") crystalline plasticizer or approved equivalent, in min. 1.75% by weight of CEM for basement tie beam, basement concrete screen wall, new basement screen wall, lift shaft, and water tank, etc.

G. The Minimum Performance of liquid applied crystalline waterproofing concrete plasticizer shall meet or excess the following requirements:

Performance	Test Methods	Minimum Requirements			
Form	-	Liquid form (powder form is not allowed)			
Technology	-	1. Functions chemically and plasticizes the concrete. 2. Nano-scale crystals are formed in the capillary system by special active catalysts, which become active on contact with water. This forms a concrete which is sustainable and permanently impermeable to water. 3. Crack healing possible for penetrating cracks up to 0.4 mm and for map / pattern cracks up to 0.5 mm. 4.			
CE Marking	EN934-2	Classify as 'Plasticizer for concrete EN 934-2: T2'			
Propagation of microorganisms on materials for drinking water	DVGW W270	Meet			
Hygienic requirements on materials for drinking water	DVGW W347	Fulfill			
Water reducing	-	Up to 10%			
The efficiency of such waterproofing admixture for concrete in comparison to a reference concrete. Compressive strength and bulk density	- DIN EN 12390-3: 2009-07 1, 7 & 28 days	- a. bulk density kg/m ³ Age Day Reference Waterproofing admixture 1 2300 2320 7 2330 2330 28 2320 2320 b. compressive strength (150mm cube) in MPa Age Day Reference Waterproofing admixture 1 16.4 20.4 7 48.5 47.5 28 57.9 58.8			
Depth of penetration of water under pressure	DIN EN 12390-8:2009-07 30 days	Reference		Waterproofing admixture	
		22mm average		10mm average	
Freeze-thaw-salt resistance	BAW-Merkblatt 28 days	a. Surface scaling in g/m ²			
		No. of Frost-thaw cycles	Reference	Water-proofing admixture	Waterproofing Admixture + air-entraining agent
		0	0	0	0
		4	36	51	59
		10	167	180	177
		14	386	345	282
		18	1053	611	376
		28	3680	1671	613
		b. Rel. Dynamic E-modulus in %			
		No. of Frost-thaw cycles	Reference	Water-proofing admixture	Waterproofing Admixture + air-entraining agent
		0	100	100	100
		4	99	99	100
		10	94	98	99
14	91	97	98		
18	85	95	98		
28	70	95	97		
Chloride migration resistance	BAW-Merkblatt 56 days	Chloride migration coefficient Meanvalue D _{CL} x 10 ⁻¹² [m²/s]			

		Reference	Waterproofing admixture
		14.7	8.9
Water Permeability Test	BTD/TP/02/2002	Average coefficient of Permeability, K (m/sec)	
		Reference	Waterproofing admixture
		2.06×10^{-12}	4.78×10^{-13}
Sulphate content (as SO ₃)	QCC/LP6-07	Less than 0.1 % (by mass of sample)	
Normatively required migration coefficients	DIN EN 206-1/ DIN 1045-2	Comply with exposure class XS1, XD1 XS2, XD2	
Service life	-	Unreacted crystalline waterproofing admixture is available to the concrete throughout its service life and continues to react on contact with water. This reaction process also works after a long period, e.g. 25 years. The life expectancy of the concrete construction is considerably increased.	

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

A. Refer to Section 1.2 Summary.

B. Hot Weather Concreting: Perform work in accordance with ACI 305R.

1. When hot weather conditions exist that would seriously impair the quality and strength of concrete, place concrete in compliance with ACI 305R and as herein specified.
2. Temperature of concrete at time of placing: Not to exceed 32.5°C. Maintain an accurate reading thermometer at the job site to check temperature of concrete. Reject concrete before placing if temperature of concrete exceeds 32.5°C.
3. Execute special precautions to protect fresh concrete before and during finishing when the rate of evaporation of surface moisture from concrete exceeds 1kg/m² per hour. Determine rate of evaporation in accordance with ACI 305R. Provide special precautions as required:
 - a. Cool ingredients before mixing to reduce concrete temperature at time of placement. Mixing clean, potable water may be chilled, or chopped ice may be used to control the concrete temperature provided the water equivalent of the ice is calculated to the total amount of mixing water.
 - b. Dampen subgrade and forms.
 - c. Cover reinforcing steel with water-soaked burlap so the steel temperature will not exceed the ambient air temperature immediately before embedment in concrete.

1.07 SUBMITTALS

A. Submit the following as required in the specification:-

1. Full Specification of the current liquid applied crystalline plasticizer admixture waterproof concrete shall be submitted to the Architect/ Engineer for approval; and concrete to be used shall be in accordance with the Drawings and current liquid applied crystalline plasticizer admixture waterproof concrete specification.
2. The Supplier shall note that holes may be left in the basement slabs for pressure relief wells during basement construction. He shall submit his proposal for sealing these holes which may be left after concreting of the basement slabs to the Architect.

B. Shop Drawings

1. All joint details and their layout (including pipes or embedment permanently or temporarily embedded in concrete work, and construction joints) in the Work shall be agreed by manufacturer and the watertightness of these joints guaranteed by the Main Contractor. The joint details and their layout shall be submitted to the Architect/Engineer for their approval. The Architect/Engineer reserves the right to alter the joint details and/or layout at any time.

C. Samples

1. The affected concrete in the structure will continue to be cured with fresh water by the use of bleeder hoses or similar system as approved by the Engineer.

2. Cores will be taken from the structure under the supervision of the Engineer. Should these core samples fail to comply with the specification, reconstruction will be necessary.
- D. Propose sub-contractor specializing in installation of cementitious systems with ten (10) years minimum experience on projects of a similar size and complexity.
- E. Where reference is made to products from various manufacturers, submit a list of the proposed products and confirm compatibility between all elements/materials as appropriate.
- F. Alternatively product from a single source supplier is permitted. Confirm compatibility between all elements/materials as appropriate.

1.08 PRE-INSTALLATION CONFERENCE

- A. Record discussions of conference and decisions and agreements reached and furnish copy of record to each party attending, and to the Architect for review. Review foreseeable methods and procedures related to the precast work, including the following:
 1. Review performance requirements (Drawings, Specifications and other Contract Documents).
 2. Review required submittals, both completed and yet to be completed.
 3. Review and finalize construction schedule related to concreting works and verify availability of materials, installer's personnel, equipment and facilities needed to make progress and avoid delays.
 4. Review required inspections, testing, certifying and material usage accounting procedures.
 5. Review weather and forecasted weather conditions, and procedures for coping with unfavorable conditions, including temporary protections against traffic.

1.09 DELIVERY STORAGE AND HANDLING

- A. Should there be a delay in the supply of concrete the Supplier of the waterproof concrete admixture will provide the Main Contractor with a surface retarder. The Main Contractor will apply the surface retarder to the affected surfaces to delay the setting time of the concrete. The surface retarder shall not have an adverse reaction on the long term performance of Waterproof Concrete System.
- B. The mixing of waterproof concrete admixture into the concrete should not be carried out on site.

1.10 MOCK-UP

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

- A. The compressive strength test is to be conducted on cores taken from cubes or cylinders as specified in the standard unless the use of cast specimens is approved by the Architect/ Engineer.
- B. Prior to construction, and at least 30 days before the actual waterproof concrete is to be cast, the Main Contractor shall arrange for trial mixes to be conducted, under the supervision of the Engineer or his representative, and with the Supplier of the waterproof concrete admixture, to confirm that the proposed mix conforms with the compressive strength requirements.

1.12 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. Provide Ten (10) years warranty for all waterproofing concreting works.
- C. The Contractor and the Applicator shall warrant that all products, materials and items are suitable, and can be maintained in the various climatic and psychometric conditions encountered without cracking, delamination, or material defects.
- D. Nothing shall be done by the Management Contractor which will void any Manufacturer's / Supplier's Warranty.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURER

- A. Subject to compliance with the Project Contract Document Requirements, provide cementitious screed and ancillary materials from the following or equivalent approved by the Architect.

1. **TECH LINK CONSTRUCTION ENGINEERING LIMITED**

Unit 802, 8/F, Laford Centre,
No. 838 Lai Chi Kok Road,
Cheung Sha Wah, Kowloon,
Tel: (852) 3701 1817 / 9316 8906
Fax: (852) 3701 1818
Email: kenuskwong@techlinkcon.com.hk; or

2. Equivalent approved by Architect

2.02 MATERIALS

A. **Waterproof Concrete**

Waterproof concrete shall be constructed using concrete with liquid applied crystalline plasticizer admixture.

PART 3 EXECUTION

3.01 PREPARATION

- A. Strictly follow the surface preparation requirements provided by the supplier of waterproof concrete admixture.
- B. The Architect/ Engineer's representative will have the right of final say in the event of disagreements.

3.02 INSTALLATION / APPLICATION

- A. Strictly follow the application procedures provided by the supplier of waterproof concrete admixture.

Instructions for Use of BETOCRETE-C17 (BV):

- i. The water-cement ratio shall not exceed 0.55. Workability time is approx. 45 minutes after the addition of BETOCRETE-C17 (BV).
 - ii. The addition of a concrete retarder may be necessary when using Portland Cement Type II or III. Suitability tests have to be carried out before usage.
 - iii. When using BETOCRETE-C17 (BV) at temperatures below +10° C, the product shall be blended with water in the ratio 1:1 prior to usage.
 - iv. If the material was stored at temperatures below +8° C crystals may form. The product is suitable for usage again after stirring or homogenizing.
- B. The Architect/ Engineer's representative will have the right of final say in the event of disagreements.
 - C. Extent of waterproofing concrete
The extent of waterproofing concrete should be for Whole Concrete Screen Wall / At least 300mm thick top of Pile Cap, Lift Pit and Water Tank. Details should be referred to the drawing.
 - D. Dosage Range:
2.0- 3.0 % by weight of cement (*), at least 7 kg per m3 (*) of concrete.
 - E. In ready-mix factory:
BETOCRETE-C17 (BV) can be dosed together with the gauging water or to the ready mix as the last component.
 - F. In truck mixer on the job site:
BETOCRETE-C17 (BV) is dosed into the mixer drum and then thoroughly mixed for 3-5 minutes. Pour without delay.

G. Placing Concrete:

1. Place concrete in accordance with the Consulting Engineers specification.
2. Ensure reinforcement, embedded parts, etc. are not disturbed during concrete placement.
3. Do not deposit concrete which has partially set, hardened or contaminated with detrimental materials.
4. Excessive honey comb or embedded debris in concrete is not acceptable.
5. Do not remove forms until concrete has attained sufficient strength. Formwork clamps or ties may be loosened 24 hours after concrete is placed.

H. Curing Interior Surfaces:

1. Cure floor surfaces in accordance with ACI 308.
2. Chemical Cure for Slabs (ACI 308): Apply in accordance with manufacturer's single coat application instructions. Obtain from the curing compound manufacturer a written warranty that the compound will not be detrimental to bonding of flooring adhesives or surface materials. Submit warranty to the Employer at the time request is made for use of curing compound. (Refer to Section 01740 - Warranties & Bonds.)
3. Moisture Cure: Moisture curing method is optional for all areas receiving special flooring (if specified) or hardener.
 - a. Place wet, moisture-containing fabric covering as soon as concrete had hardened sufficiently to prevent surface damage.
 - b. Cover entire surface, including edges and sides.
 - c. Keep coverings continually moist so that a film of water remains on concrete surface throughout curing period. Maintain concrete in moist condition for not less than seven days after placement.

3.03 CLEANING

(NOT USED)

3.04 PROTECTION

(NOT USED)

END OF SECTION